

PLEVEN, Bulgaria

WMO: 155280

Lat: 43.400N

Lon: 24.600E

Elev: 525

StdP: 14.42

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.7	14.8	5.1	7.3	10.6	10.8	9.8	16.3	32.3	42.4	27.5	42.2	2.3	70	0.445

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	94.4	71.7	91.1	71.2	88.2	70.2	74.8	88.8	73.0	86.7	71.5	84.8	4.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.3	113.6	83.1	68.4	106.3	80.7	66.8	100.5	78.7	38.6	89.3	37.0	86.8	35.6	84.9	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.3	22.2	17.3	DB	2.5	99.7	6.2	4.0	-1.9	102.6	-5.5	104.9	-9.0	107.1	-13.4	110.0	
			WB	2.0	78.3	6.2	3.3	-2.5	80.7	-6.1	82.6	-9.5	84.4	-14.0	86.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.5	32.4	36.0	45.2	55.0	64.5	71.2	75.0	74.7	66.0	55.0	44.0	33.4	(6)
(7)		DBStd	17.11	9.49	10.65	8.72	6.55	6.17	5.66	5.10	5.34	6.31	7.92	8.09	9.02	(7)
(8)		HDD50	1957	550	398	201	29	1	0	0	0	42	216	520	(8)	
(9)		HDD65	4839	1011	811	616	305	85	13	2	3	62	319	631	981	(9)
(10)		CDD50	3585	3	7	50	178	451	635	775	767	481	198	36	4	(10)
(11)		CDD65	995	0	0	0	4	71	198	312	305	93	11	1	0	(11)
(12)		CDH74	9730	0	0	6	73	656	1886	3149	3007	839	111	3	0	(12)
(13)		CDH80	3925	0	0	0	14	161	712	1408	1339	270	21	0	0	(13)
(14)	Wind	WSAvg	5.2	5.0	6.0	7.1	6.0	5.2	5.1	4.8	4.4	4.5	4.3	4.8	4.6	(14)
(15)	Precipitation	PrecAvg	23.0	1.7	1.5	1.5	2.0	2.6	3.1	2.4	1.9	1.6	1.3	1.9	1.7	(15)
(16)		PrecMax	29.8	4.5	2.9	4.8	4.5	5.4	6.9	5.5	5.9	6.1	5.9	4.2	3.7	(16)
(17)		PrecMin	15.9	0.0	0.1	0.1	0.0	0.2	0.6	0.2	0.1	0.1	0.0	0.0	0.1	(17)
(18)		PrecStd	4.2	1.1	0.7	1.1	1.1	1.2	1.8	1.3	1.5	1.5	1.2	1.1	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.7	66.2	73.8	81.6	88.8	95.0	99.1	99.5	92.6	82.7	71.8	61.8	(19)
(20)		MCWB	48.6	53.3	54.5	62.7	69.0	72.5	71.2	71.0	69.7	66.4	58.7	52.0	(20)	
(21)		2%	DB	53.9	60.7	68.7	74.7	84.3	90.7	94.6	94.3	86.7	76.8	65.8	54.5	(21)
(22)		MCWB	45.3	49.9	53.5	59.9	66.5	71.7	71.4	71.8	67.3	63.2	56.3	48.2	(22)	
(23)		5%	DB	49.4	56.1	64.2	70.7	80.9	87.4	90.9	90.8	82.6	72.1	59.9	48.7	(23)
(24)		MCWB	43.7	47.3	50.9	57.2	65.2	71.0	71.4	70.8	65.4	60.9	52.7	44.3	(24)	
(25)		10%	DB	45.1	50.8	59.6	67.4	77.5	84.1	87.8	87.5	78.6	67.8	55.6	44.5	(25)
(26)		MCWB	40.9	44.6	49.0	55.1	63.9	69.4	70.4	69.5	63.8	58.8	50.5	41.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.3	54.0	57.6	64.5	70.9	76.3	77.3	78.6	72.3	68.4	60.1	52.4	(27)
(28)		MCDB	57.5	63.5	69.2	79.6	85.4	88.9	90.8	90.8	87.1	80.7	68.3	58.9	(28)	
(29)		2%	WB	46.3	50.5	54.4	61.0	68.6	73.7	74.5	74.8	69.1	64.2	56.6	48.4	(29)
(30)		MCDB	52.4	59.8	65.9	72.7	81.3	87.1	88.8	89.0	83.3	74.4	64.5	54.0	(30)	
(31)		5%	WB	43.9	47.6	51.9	58.5	66.6	72.0	72.8	72.6	67.0	61.5	54.0	44.9	(31)
(32)		MCDB	49.0	55.0	62.3	68.5	78.1	84.9	87.0	85.4	79.7	70.6	58.6	48.6	(32)	
(33)		10%	WB	41.1	44.7	49.5	56.3	64.7	70.2	71.3	70.6	65.1	59.2	51.3	42.0	(33)
(34)		MCDB	45.1	50.4	59.2	65.4	75.6	82.2	84.9	83.8	76.2	66.7	55.2	44.4	(34)	
(35)	Mean Daily Temperature Range	MDBR	11.1	14.3	16.9	18.5	19.7	20.2	20.9	21.3	19.5	17.2	12.8	10.7	(35)	
(36)		5% DB	MCDBR	18.0	24.3	25.0	25.3	25.5	25.7	26.7	27.2	26.6	24.8	21.3	17.9	(36)
(37)		MCWBR	12.1	15.4	13.9	13.4	12.0	11.3	10.8	10.3	11.4	13.5	13.2	12.1	(37)	
(38)		5% WB	MCDBR	17.3	23.2	24.1	23.3	23.3	24.1	24.0	24.1	24.5	23.4	19.2	16.4	(38)
(39)		MCWBR	12.2	15.3	14.7	13.6	12.7	12.3	11.7	11.8	12.4	13.6	12.6	11.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.337	0.367	0.408	0.464	0.443	0.447	0.456	0.467	0.436	0.395	0.362	0.341	(40)	
(41)		taud	2.314	2.241	2.179	2.034	2.165	2.193	2.173	2.139	2.197	2.296	2.353	2.333	(41)	
(42)		Ebn at Noon	246	257	261	255	264	263	259	251	249	244	235	232	(42)	
(43)		Edh at Noon	27	34	41	51	46	45	46	46	40	32	26	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	470	734	1096	1501	1843	2034	2063	1846	1345	878	508	390	(44)	
(45)		RadStd	71	94	106	155	153	129	126	123	104	105	69	53	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	-1.31	N/A	N/A	N/A	N/A	(46)
(47)	Regional (48 neighbors)		+0.85	N/A	N/A	+0.76	N/A	N/A	-256	+126	(47)

Nomenclature: See separate page